

illustrated in the figure itself. Then we sum the group lasso over all groups as a regulation and add it as a part of the optimisation terms of the overall optimisation objectives. This forms the procedure of our method called the Structure Sparsity Learning.

Sparsity-inducing regulariser for DNN involves the following:

L1 regulariser (sum of absolute values)

- It has been used for sparsity since 1996.
 - It is differentiable, convex, and easy to optimise
 - It is proportional to the scaling, which will lead to scaling down of all the elements with the same speed, which is undesired
- L0 regulariser (number of nonzero values)*
- It directly reflects the sparsity by definition
 - It is scale invariant. However, our zero regulariser provides no useful gradients
 - It needs additional tricks for applying on DNN pruning (stochastic approximation, ADMM), which makes the problem complicated. This leads to certain complexity in the design

Moving beyond lasso

We propose to move beyond L1 and L0 and aim to find the sparsity-inducing regulariser that is both differentiable and scale-invariant for pruning. At this stage the hoyer-square regulariser draws our attention. It is represented by the ratio between the L1 and L2, which is both differentiable and scale-invariant.

For element-wise DNN pruning, we can apply the square version of hoyer regulariser, which has the same range and similar minima structure as L0. We further extend the hoyer-square regulariser to the structure pruning settings.

We can apply hoyer-square regulariser over the L2 norm of each group of weight. The weight within the group can be induced to become zero simultaneously. Let us now look at the effect of the hoyer-square regulariser.

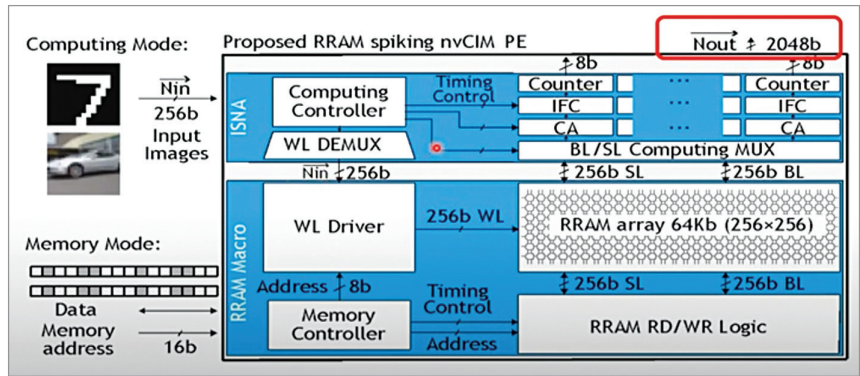


Fig. 8: Processing engine with in-situ neuron activation

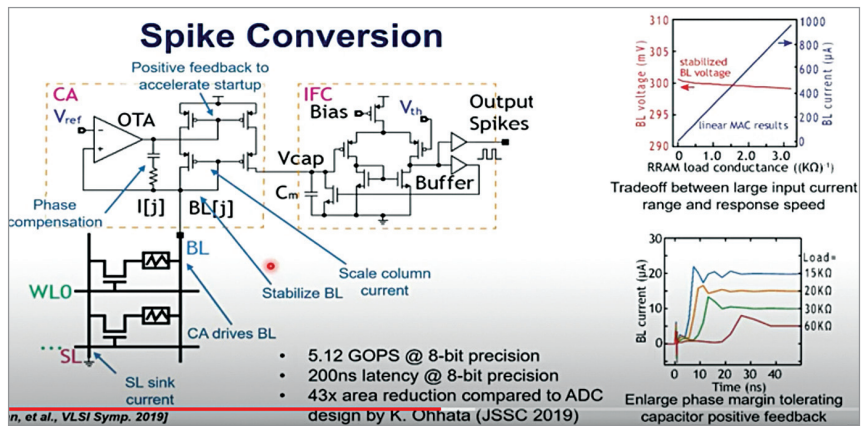


Fig. 9: Spike conversion

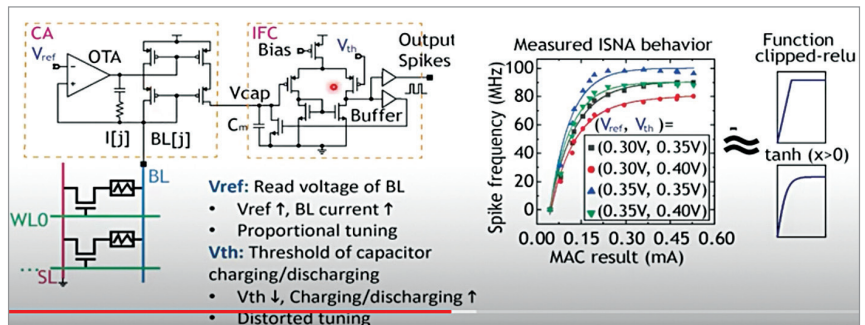


Fig. 10: In-situ nonlinear activation function

Regulating the activation sparsity. In operations, the activation sparsity is dynamic and the sparsity level is not identical for different inputs. Our group has proposed what is called the Dynamic Winner Takes All (WTA) dropout technique to prune neuron activations layer wise. Hence, the design introduces low overhead for the real-time interface (see Fig. 5).

Mixing precision with bit-level sparsity. We know that fixed point quantisation is an important model

compression technique. In the previous research it was found that some layers can retain higher precision while others can be quantised to less precision. This is the whole idea of mixed precision quantisation.

For a quantised matrix, when the most significant bits of all the elements are zero simultaneously, we can remove the MSB directly (see Fig. 6).

For bit representation we follow the dynamic scaling quantisation procedure. Here (see Fig. 7) we use the ele-